

# MUHAMMAD HAMZA NIAZ

King of Prussia, PA | (929) 354-5032 | hamzaniaz9856@gmail.com|  
linkedin.com/in/muhammad-n/| github.com/AlphaPro06|US Citizen

## Experience

### University at Buffalo, State University of New York

#### Research Assistant

Buffalo, NY

June 2025 – Present

- Creating interactive data visualizations and comprehensive analytical reports to communicate research findings to academic stakeholders, department leadership, and potential funding organizations
- Applying advanced statistical methods including regression analysis and hypothesis testing to support peer-reviewed research publications and conference presentations
- Collaborating with interdisciplinary research teams to ensure data integrity, validate findings, and maintain research compliance standards
- **Tech Stack:** Python, Statistical Analysis, Data Visualization, Research Methodology, Academic Research Tools

### The Vanguard Group, Inc

#### Application Engineer

Malvern, PA

July 2023 – March 2025

- Built AWS Lambda functions and Glue jobs in Python to integrate Bloomberg PORT data with AWS Athena, enabling fund managers to perform comparative risk analysis between PORT and Aladdin models
- Developed SQL queries and data integration workflows to migrate and synchronize Derivatives and Fund-of-funds data between internal MADB database and MSCI platform, ensuring data accuracy and consistency
- Maintained and monitored 5 enterprise financial data processing pipelines running daily, handling gigabytes of financial data and achieving 99.8% system availability through proactive incident management
- Developed Python automation scripts for data transformation workflows, converting and validating financial data between vendor formats and internal database schemas for insert/upsert operations
- Led 6-month migration of 10 team repositories from Atlassian Bitbucket to GitHub for a cross-functional team of 10 members, implementing GitHub Actions workflows with integrated Checkmarx security scanning
- Resolved 10-15 production incidents monthly, troubleshooting data pipeline failures, system errors, and data quality issues while meeting SLA commitments for critical financial processing systems
- Collaborated with business stakeholders including portfolio managers and risk analysts to gather requirements and resolve production data processing issues
- **Tech Stack:** Python, AWS (S3, IAM, Lambda, CloudWatch, CloudFormation, Step Functions), GitHub Actions, SFTP, SQL (PostgreSQL), NumPy, Excel, Pandas, Tableau

### Accenture

#### Software Engineering Summer Analyst

New York, NY

June 2022 – August 2022

- Streamlined 30% reduction in manual testing effort as measured by deployment cycle time by developing automated cloud deployment scripts using AWS Lambda and HTML, CSS, JavaScript automation tools
- Developed responsive UI components using JavaScript and React, implementing A/B testing methodologies to evaluate and optimize user experience and engagement metrics
- Built an Alexa voice bot integrated with AWS Lambda and Splunk API to enable rapid service outage reporting, designing front-end dashboards with real-time system status visualization
- **Tech Stack:** JavaScript, TypeScript, React, AWS (Lambda), Splunk API, Excel (advanced functions, pivot tables, macros), Web Application Testing, SQL

### Research Foundation for The State University of New York

#### Research Aide

Buffalo, NY

June 2021 – August 2021

- Analyzed 15,000+ multidimensional data points as measured by dataset size and complexity by developing responsive web application with interactive statistical graphs visualizing scientific datasets for correlation identification
- Validated statistical significance as measured by hypothesis testing results by applying advanced statistical methods and data modeling techniques to complex research datasets
- Processed 3 terabytes of research data as measured by data volume throughput by utilizing Python data visualization libraries to translate mathematical concepts into practical applications
- **Tech Stack:** Flask, JavaScript, HTML5, CSS3, Python data visualization libraries

## Skills

**Languages & Frameworks:** Python, Java, JavaScript, TypeScript, React.js, Node.js, Spring Boot, Pytest, Cucumber, C/C++, SQL

**Cloud & Tools:** AWS, Git, GitHub Actions, Bamboo, Splunk Databases

**Analytics:** DynamoDB, PostgreSQL, MongoDB, Tableau, SQL Server

## Education

### University at Buffalo, State University of New York

#### Bachelor of Science in Computer Science

Buffalo, NY

Graduated: May 2023

**Relevant Coursework:** Data Structures, Algorithms, Systems Programming, Linear Algebra, Probability Theory, Computer Vision and Image Processing, Applied Human-Computer Interaction, Modern Networking Concepts, Distributed Systems, Software Engineering, Web Applications, Data-Intensive Computing